

LINGJUN ZHAO

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SUMMARY

Research area: Natural language processing (NLP) and generation (NLG), human-AI collaboration.
Topics: Visual-language model, pragmatics, explainability, uncertainty, alignment.

EDUCATION

University of Maryland, College Park Jan 2021 - Present
Ph.D. candidate in Computer Science
Advisor: Prof. Hal Daumé III
Thesis: Methods for Improving the Communication Efficacy of LLM: Faithfulness and Pragmatics
Courses: Human-AI Interaction, AI Agents, Computational Linguistics, Foundation of Deep Learning

Columbia University, New York Sep 2016 - Feb 2018
M.S. in Computer Science. Track: Machine learning

Sun Yat-Sen University, Guangzhou, China Sep 2012 - Jun 2016
B.S. in Computational Mathematics

PREPRINTS

P1 **Pragmatics Meets Culture: Culturally-adapted Artwork Description Generation and Evaluation** [\[paper\]](#)
L. Zhao, D. Ki, M. Carpuat, H. Daumé III
In submission 2026

CONFERENCE PAPERS

P2 **A Necessary Step toward Faithfulness: Measuring and Improving Consistency in Free-Text Explanations** [\[paper\]](#) [\[code\]](#)
L. Zhao, H. Daumé III
Empirical Methods in Natural Language Processing (EMNLP) 2025

P3 **Can Hallucination Correction Improve Video-Language Alignment?** [\[paper\]](#)
L. Zhao, M. Xie, P. Cascante, H. Daumé III, K. Lee
Findings of Association for Computational Linguistics (ACL) 2025

P4 **Successfully Guiding Humans with Imperfect Instructions by Highlighting Potential Errors and Suggesting Alternatives** [\[paper\]](#) [\[code\]](#)
L. Zhao, K. Nguyen, H. Daumé III
EMNLP, Oral 2024

P5 **Hallucination Detection for Grounded Instruction Generation** [\[paper\]](#) [\[code\]](#)
L. Zhao, K. Nguyen, H. Daumé III.
Findings of EMNLP 2023

P6 **Define, Evaluate, and Improve Task-Oriented Cognitive Capabilities for Instruction Generation Models** [\[paper\]](#) [\[code\]](#)
L. Zhao, K. Nguyen, H. Daumé III
Findings of ACL 2023
Also presented at ICML 2023 Theory-of-mind Workshop: Outstanding Paper Award

- P7 **Towards Few-Shot Event Mention Retrieval: An Evaluation Framework and A Siamese Network Approach** [\[paper\]](#)
 B. Min, Y. Chan, **L. Zhao**
 Language Resources and Evaluation Conference (LREC) 2020
- P8 **Neural-Network Lexical Translation for Cross-lingual IR from Text and Speech** [\[paper\]](#)
 R. Zbib, **L. Zhao**, ..., J. Makhoul
 ACM Special Interest Group in Information Retrieval (SIGIR) 2019

WORKSHOP & JOURNAL PAPERS

- WJ1 **Leveraging Computer Vision for Predicting Collision Risks** [\[paper\]](#)
 Q. Nguyen, ..., **L. Zhao**, ..., T. Tasdizen
 Injury Prevention Journal 2025
- WJ2 **Cross-lingual Information Retrieval with BERT** [\[paper\]](#)
 Z. Jiang, Z. El-Jaroudi, W. Hartmann, D. Karakos, **L. Zhao**
 LREC Workshop 2020
- WJ3 **The 2019 BBN Cross-lingual Information Retrieval System** [\[paper\]](#)
 L. Zhang, ..., **L. Zhao**, ..., J. Makhoul
 LREC Workshop 2020
- WJ4 **Weakly Supervised Attentional Model for Low Resource Cross-lingual Information Retrieval** [\[paper\]](#)
L. Zhao, R. Zbib, Z. Jiang, D. Karakos, Z. Huang
 EMNLP Deep Learning Approaches for Low-Resource NLP Workshop 2019

RESEARCH EXPERIENCES

Graduate Research Assistant Jan 2022 - Present
University of Maryland College Park, MD

Develop methods to improve human-AI communication for human-AI collaboration:

- Improve pragmatic communication through human behavior simulation using RL agents, or visual-language model self-improvement [\[P1, P6\]](#)
- Measure and improve explanation consistency towards faithfulness [\[P2\]](#)
- Rich uncertainty generation and communication to improve human task performance [\[P4, P5\]](#)

Research Intern Jun 2025 - Aug 2025
Microsoft Research, Copilot Tuning Team (formerly Semantic Machines) Baltimore, MD

- Fine-tuned models with direct preference optimization to personalize to users' writing styles using limited training samples, improved over strong baseline by 5%. Research integrated into product

Research Intern May 2024 - Aug 2024
Honda Research Institute San Jose, CA

- Proposed methods and fine-tuned models for video-language alignment, improving performance by up to 18% [\[P3\]](#)

Staff Scientist II Mar 2018 - Dec 2020
Raytheon BBN Technologies Cambridge, MA

- Developed models for low resource cross-lingual information retrieval, achieved state-of-the-art in the IARPA MATERIAL program [\[P8, WJ4, WJ2, WJ3\]](#)
- Developed event retrieval model for the IARPA BETTER program [\[P7\]](#)

AWARDS

- **Travel Award**, University of Maryland 2024
- **Outstanding Paper Award**, ICML Theory-of-Mind Workshop 2023
- **First Prize (top 3%)**, ASC Student Supercomputer Challenge 2015
- **Outstanding Student Scholarship**, Sun Yat-Sen University 2015
- **Second Place**, Mathematical Contest in Modelling, Sun Yat-Sen University 2012

TALKS

- *Methods for Improving the Communication Efficacy of Language Models*, Google DeepMind 2025
- *Successfully Guiding Humans with Imperfect Instructions by Highlighting Potential Errors and Suggesting Alternatives*, EMNLP conference 2024
- *Define, Evaluate, and Improve Task-Oriented Cognitive Capabilities for Instruction Generation Models*, ICML Theory-of-Mind workshop 2023
- *Bridging Human-AI Communication Gap for Visual-language Navigation*, University of Maryland Human-Computer Interaction Lab Symposium 2023
- *Generating and Evaluating Visually Grounded Navigation Instruction*, Mid-Atlantic Student Colloquium on Speech, Language and Learning 2022

TEACHING

- *Teaching assistant*, courses: AI agents, Human-AI interaction, Intro to machine learning 2021-2025
- *Mentor for underrepresented high school students*, TRAILS AI Summer Academy, U Maryland 2023

SERVICES

- *Conference reviewer*: ACL, EMNLP 2024-Present
- *Student volunteer*, University of Maryland graduate admissions committee 2021
- *R&D director*, Sun Yat-Sen University Computational Science Association 2014-2015